COLLEGE BLOCKCHAIN PROJECT / DAPP

SkillProof

Trust skills. Verify credentials.
Powered by blockchain.



01

The trust gap in credentials

Traditional certificates are easy to claim, slow to validate, and dependent on fragmented issuer records.

Easy to alter

Files and screenshots can be edited or presented without authoritative proof.

Hard to verify

Manual checks require emails, portals, and institutional response time.

No shared truth

Credentials remain siloed across institutions and disconnected systems.

02

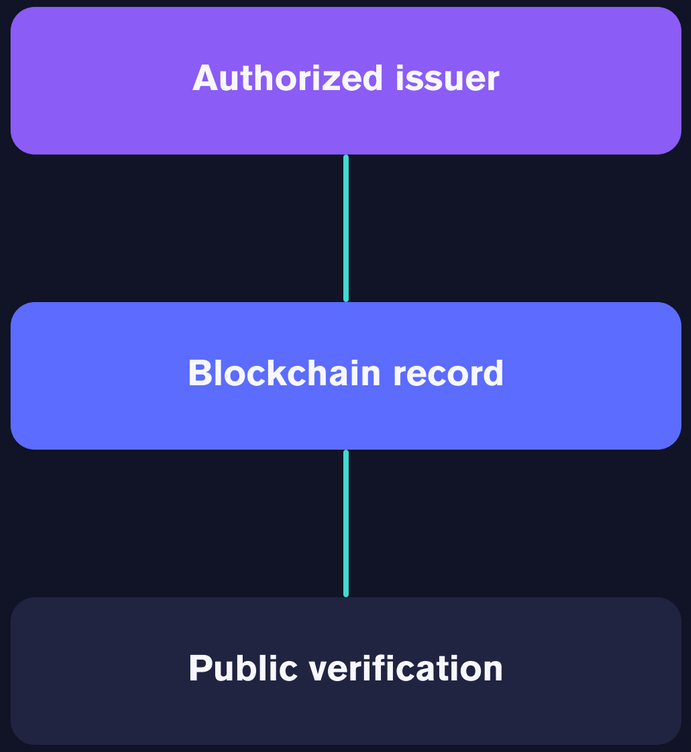
A credential with built-in proof

SkillProof anchors issuer-authorized credential records on-chain and gives anyone a direct verification path.

VERIFIED CREDENTIAL

Blockchain Development

Issued to a wallet address
Immutable credential ID • Active status



03

From issuance to instant verification

1 Issuer creates

Authorized account submits credential data.

2 Recipient receives

Credential ID is shared with the learner.

3 Blockchain anchors

Contract stores issuer, recipient and status.

4 Verifier checks

Public lookup returns trusted state.

04

Core capabilities, implemented

Issue

Create wallet-linked credentials

Verify

Look up by credential ID

Revoke

Invalidate compromised claims

Track

Read issuer and recipient data

Connect

Use a browser wallet

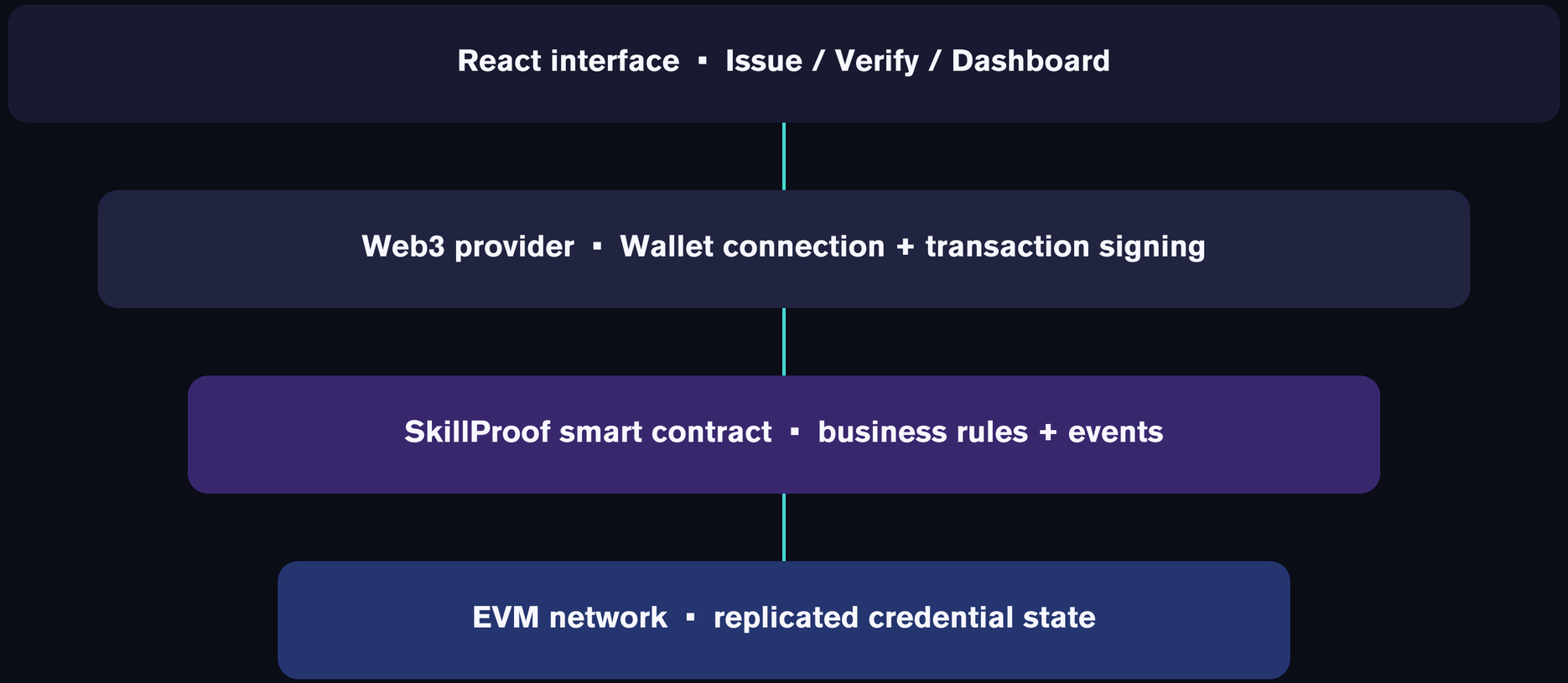
Surface states

Valid, revoked, or not found

05

System architecture

A deliberately small, auditable stack from user action to immutable state.



06

The contract is the source of truth

The deployed SkillProof contract controls who can write credentials and how their lifecycle changes.

Access control

Restricts credential issuance and revocation to authorized issuer accounts.

Credential registry

Stores credential ID, issuer, recipient, metadata reference, timestamp, and status.

Events + reads

Emits lifecycle events and exposes public functions for deterministic verification.

07

Frontend pages with one clear job

Home

Explain the product and route users

Issue

Create and submit a credential

Verify

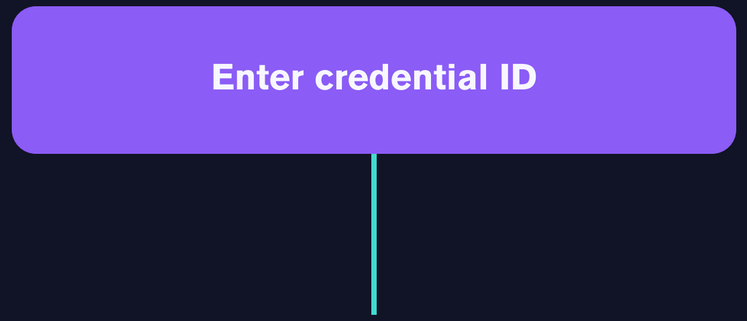
Search and interpret credential state

Dashboard

Review issued credentials

08

One lookup. Three explicit outcomes.



VALID

Issuer and recipient match; credential remains active.

REVOKED

Record exists, but the issuer has invalidated it.

NOT FOUND

No credential record exists for the submitted ID.

09

Technology stack

Familiar web tooling meets an auditable smart-contract layer.

UI

React • JavaScript • CSS

Web3

ethers.js • browser wallet

Contract

Solidity • access control • events

Dev

Hardhat • local tests • deployment scripts

Network

EVM-compatible blockchain

Workflow

Git • GitHub • environment config

What the prototype proves

SkillProof demonstrates a complete credential lifecycle rather than a disconnected smart-contract demo.

ISSUE → VERIFY → REVOKE

End to end

UI actions map to real contract transactions.

Transparent

Credential state is publicly readable and consistent.

Portfolio ready

A focused foundation for demos, review, and extension.

Practical next steps

These are future directions—not claims about the current prototype.

01

Decentralized metadata

Store richer credential evidence with content-addressed files.

02

Issuer governance

Add multi-organization roles, onboarding, and policy controls.

03

Shareable proofs

Generate QR-based verification links and portable credential views.

04

Production readiness

Expand testing, monitoring, network deployment, and security review.

Thank you.

SkillProof / Blockchain credential verification